Data-report

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0.1 Report: Analyzing the relations between the renewable energy consumption and Co2 emission .

0.1.1 Main Question

Does consumption of renewable energy sources decreases the Co2 emission?

0.1.2 Data Sources

To answer the question, two data sources have been selected for this project: The world Bank, which shows yearly information on Latin Americans renewable energy consumption, and Our World in Data, which shows the Co2 emissions from Latin American countries.

Data source 1: Renewable Energy Consumption

- **Metadata URL**: Renewable energy consumption (% of total final energy consumption) | Data
- **Data URL**: https://api.worldbank.org/v2/en/indicator/EG.FEC.RNEW.ZS?downloadformat=csv
- **Data Type**: CSV Directory
- **Description**: This dataset from the World Bank tracks the share of renewable energy in total final energy consumption, presented as a percentage. It highlights global and regional trends in transitioning to renewable energy sources. Includes country-level data on renewable energy consumption as a percentage of total energy use from 1960 to recent years.
- **Why Chosen**: This dataset provides critical insights into renewable energy adoption, essential for analyzing sustainability efforts and energy policy effectiveness.
- **Data Structure**: Organized as a time series, it includes columns for year, country, and renewable energy's percentage share in energy consumption.
- **Data Quality**: Sourced from the World Bank, it is generally reliable but may have gaps or variations in accuracy due to regional reporting standards.
- **License and Obligations**: The dataset from the World Bank is typically shared under

the **Creative Commons Attribution 4.0 International (CC BY 4.0)** license. This allows for use, sharing, and adaptation of the data, provided proper attribution is given to the world bank.

- **License Source: Data Access And Licensing | Data Catalog**

Data source 2: Co2 emissions Data

- **Metadata URL**: https://data.worldbank.org/indicator/EN.GHG.CO2.PI.MT.CE.AR5

- **Data URL**:

  https://api.worldbank.org/v2/en/indicator/EN.GHG.CO2.PI.MT.CE.AR5?downloadformat=csv

- **Data Type**: CSV

- **Description**: This dataset provides per capita CO_2 emissions by Latin Americans region, showcasing historical trends in carbon dioxide emissions across different geographical areas. It helps in understanding regional contributions to global emissions and monitoring progress toward climate goals. Regional data on CO_2 emissions measured in metric tons per capita over multiple decades.

- **Data Structure**: The dataset is organized as a time series with columns for year, region, and per capita CO_2 emissions measured in metric tons. Each record corresponds to a specific region and year, facilitating temporal and spatial analysis.

- **Data Quality:** The data is sourced from reputable research institutions and may vary in accuracy due to regional differences in reporting standards and estimation methods. Efforts have been made to standardize the data, but gaps or approximations may exist for certain regions or time periods.

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0.1.2.1 Data Pipeline

The pipeline processes datasets from multiple sources, such as World Bank and Our World in Data, focusing on renewable energy consumption and CO_2 emissions data. It uses Python for data extraction, transformation, and storage, employing libraries like pandas, requests, and sqlite3.

Technologies:

- **Data Extraction:** requests for downloading ZIP files**.**
- **Transformation:** pandas for cleaning, reshaping, and filtering.
- **Storage:** CSV files and SQLite for structured data persistence.

Cleaning and Transformation Steps:

- **Filtering:** Selected North American countries and relevant years (2016–2022) to focus on regional analysis**.**
- **Reshaping**: Converted wide-format data into a long-format table for easier analysis.
- **Handling Missing Data:** Removed entries with missing or incomplete values to improve data quality.
- **Standardization:** Harmonized country names and column structures across datasets.

Meta-Quality Measures:

- **Validation**: Checked for null values, duplicate rows, and schema consistency after each step.
- **Error Handling**: Incorporated exception handling during file downloads and database writes to ensure pipeline robustness.
- **Dynamic Inputs**: Adapted to changing data formats by making the column selection and processing flexible.

0.1.3 Results And Limitations

Renewable energy consumption Dataset:

Country Name	Country Code	Indicator Name	Year	Value
Cuba	CUB	Renewable energy consumption (% of total final energy...	*2017*	*22*
Mexico	MEX	Renewable energy consumption (% of total final energy...	*2017*	*10.1*
Cuba	CUB	Renewable energy consumption (% of total final energy...	*2018*	*24*
Mexico	MEX	Renewable energy consumption (% of total final energy...	*2018*	*10.3*

Co2 emission Dataset:

Country Name	Country Code	Indicator Name	Year	Value
Cuba	CUB	Carbon dioxide (CO2) emissions from Power Industry (E...	*2017*	*14.1482*
Mexico	MEX	Carbon dioxide (CO2) emissions from Power Industry (E...	*2017*	*153.7196*
Cuba	CUB	Carbon dioxide (CO2) emissions from Power Industry (E...	*2018*	*14.2416*
Mexico	MEX	Carbon dioxide (CO2) emissions from Power Industry (E...	*2018*	*142.3862*

Output Data:

- **Structure**: Long-format tables with columns for country, year, and metric value (e.g., CO_2 emissions or renewable energy consumption).
- **Quality**: High quality but limited by missing data for certain countries or years.

Output Format:

- **CSV**: Chosen for its simplicity, portability, and compatibility with visualization tools.
- **SQLite**: Used for structured storage to facilitate complex queries during analysis.

Critical Reflection:

- **Strengths**: Provides clean, standardized data ready for analysis, focusing on North America.
- **Limitations**: Excludes countries with insufficient data and only covers recent years, potentially overlooking long-term trends.
- **Anticipated Issues**: Incomplete records for small countries or regions could bias analysis. Furthermore, variations in data collection methodologies across sources might impact consistency.